The Virasoro spectral R-matrix: closed form and non-formality

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Abstract

The spectral R-matrix of the Virasoro algebra on a primary state of conformal weight h admits the closed form $R(z) = z^{2h}\exp(-c/(4z^2))$. Only even powers of $1/z$ appear, reflecting bosonic parity. The non-terminating character of the exponential series is the genus-0 signature of shadow class M: unlike Heisenberg ($R(z) = z^k$, polynomial) or affine Kac–Moody ($R(z) = z^{\Omega/(k+h^\vee)}$, terminating), the Virasoro R-matrix has infinitely many corrections in $1/z^2$. The cubic shadow coefficient $S_3(\mathrm{Vir}_c) = 2$ is independent of the central charge c, providing a finite algebraic proof of class M membership. The first correction $R_2 = -c/4$ is the genus-0 non-formality witness from which the infinite primitive-cumulant series descends.

1 Introduction

Let $\mathcal{A}$ be a chiral algebra on a smooth projective curve X. The bar construction $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ carries a Maurer–Cartan element $\Theta_{\mathcal{A}}$ in the modular convolution dg Lie algebra. The binary genus-0 collision residue

$$r(z) := \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}) \tag{1.1}$$

is the spectral r-matrix of $\mathcal{A}$. The collision residue lives on the ordered bar complex $B^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with deconcatenation coproduct; it is an element of the E_1 convolution algebra $\mathfrak{g}^{E_1}_{\mathcal{A}}$ that retains the full spectral-parameter dependence. The modular characteristic $\kappa(\mathcal{A}) = \mathrm{av}(r(z))$ is its Σ_2-coinvariant.

The r-matrix encodes the leading singularity of the bar differential and controls the commuting Hamiltonians of the shadow connection at genus 0. On a rank-1 primary line, the path-ordered exponential of $r(z)$ produces the spectral R-matrix, which is a scalar function of the spectral parameter z.

For the Virasoro algebra Vir_c, the R-matrix on a primary state of weight h closes into an elementary function. This paper derives the closed form, expands it in series, extracts the shadow coefficients, and uses the result to distinguish the Virasoro from simpler families at the genus-0 level.

The passage from OPE to r-matrix uses the $d\log$ absorption principle: the bar propagator $d\log(z-w) = dz/(z-w)$ absorbs one power of the OPE singularity, so the collision residue has pole orders one less than the OPE. The Virasoro OPE

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w} \tag{1.2}$$

has poles at orders $4, 2, 1$. After $d\log$ absorption, the collision residue carries poles at orders $3, 1$:

$$r^{\mathrm{coll}}(z) = \frac{c/2}{z^3} + \frac{2T}{z}. \tag{1.3}$$

The simple OPE pole (∂T term) is absorbed entirely and contributes nothing to the collision residue. The even OPE pole at $(z-w)^{-2}$ becomes an odd r-matrix pole at z^{-1}, and the fourth-order OPE pole becomes a third-order r-matrix pole. For a single-generator bosonic algebra, the $d\log$ absorption sends z^{-2n} to $z^{-(2n-1)}$: only odd-order poles survive in the r-matrix.

2 The closed form

Theorem 2.1 (Virasoro spectral R-matrix on primary states)**.** *Let* Vir_c *be the Virasoro vertex algebra at central charge* c *in BPZ normalisation (so* $T(z)T(w) \sim (c/2)(z-w)^{-4} + 2T(w)(z-w)^{-2} + \partial T(w)(z-w)^{-1}$*, with the fourth-order pole coefficient* $c/2$*). For* Vir_c *acting on a primary state* $|h\rangle$ *of conformal weight* h*, the spectral* R*-matrix is*

$$R(z) = z^{2h}\,\exp\Big(-\frac{c}{4z^2}\Big) = z^{2h}\sum_{k=0}^{\infty}\frac{(-c/4)^k}{k!\,z^{2k}}. \tag{2.1}$$

The coefficient $-c/4$ *is uniquely determined: the BPZ normalisation fixes the fourth-order OPE pole at* $c/2$*;* $d\log$ *absorption sends* $(z-w)^{-4}$ *to the* r*-matrix pole* $(c/2)/z^3$ *(equation (1.3)); contour integration* $\oint -(c/2)\,z^{-3}\,dz = (c/2)\cdot(-1/(2z^2)) = -c/(4z^2)$ *produces the exponent. In particular the pole-3 datum* $S_3(\mathrm{Vir}_c) = 2$ *is fixed by BPZ, and there is no residual scaling freedom compatible with the* $d\log$*-regularised OPE at* $c \neq 0$*.*

Proof. On a primary state $|h\rangle$, the Virasoro modes act as $L_0|h\rangle = h|h\rangle$ and $L_m|h\rangle = 0$ for $m \geq 1$. The collision residue (1.3) has two independent generators on the primary sector: the identity $\mathbf{1}$ (from the central term $c/2$) and L_0 (from the linear term $2T$). Since $\mathbf{1}$ and L_0 commute on the primary line, the path-ordered exponential reduces to the ordinary exponential:

$$R(z)\,|h\rangle = \exp\left(\oint \sum_{n\geq 0} r_n\, z^{-n-1}\, dz\right)|h\rangle = \exp\left(2h\log z - \frac{c/2}{2z^2}\right)|h\rangle. \quad (2.2)$$

The $2h\log z$ comes from integrating $2h/z$ (the L_0 eigenvalue multiplied by the simple-pole coefficient in the r-matrix). The $-c/(4z^2)$ comes from integrating $-(c/2)/z^3$ along the contour. Exponentiating gives $R(z) = z^{2h}\exp(-c/(4z^2))$. Uniqueness of the coefficient $-c/4$ follows from the BPZ hypothesis: any rescaling $T \mapsto \alpha T$ rescales the fourth-order pole to $\alpha^2 c/2$, which contradicts BPZ normalisation unless $\alpha = 1$; hence the exponent $-c/(4z^2)$ is an invariant of $(\mathrm{Vir}_c, \mathrm{BPZ})$. □

The first four series coefficients are

$$R_0 = z^{2h}, \qquad R_2 = -\frac{c}{4}, \qquad R_4 = \frac{c^2}{32}, \qquad R_6 = -\frac{c^3}{384}. \quad (2.3)$$

3 Series expansion and bosonic parity

The exponential $\exp(-c/(4z^2))$ is a power series in z^{-2} with alternating signs:

$$\exp\left(-\frac{c}{4z^2}\right) = 1 - \frac{c}{4z^2} + \frac{c^2}{32z^4} - \frac{c^3}{384z^6} + \cdots = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!}\left(\frac{c}{4}\right)^k z^{-2k}. \quad (3.1)$$

Only even powers of $1/z$ appear. This is the bosonic parity constraint: for a single-generator bosonic algebra, the $d\log$ absorption sends the OPE poles at even orders z^{-2n} to odd r-matrix poles $z^{-(2n-1)}$, and the path-ordered exponential preserves the parity of the exponent. The R-matrix has the form $z^{2h}\cdot f(z^{-2})$ where f is an entire function of z^{-2}.

Remark 3.1 (Radius of convergence). The series (3.1) converges absolutely for all $z \neq 0$. The R-matrix $R(z) = z^{2h}\exp(-c/(4z^2))$ is an entire function of z^{-2} multiplied by the monomial z^{2h}. It has an essential singularity at $z = 0$: the non-terminating character of the series is the datum that distinguishes class M from classes G, L, and C, where the corresponding R-matrix is polynomial or rational in z.

Remark 3.2 (The k-th coefficient). The general coefficient in the Laurent expansion of $R(z)/z^{2h}$ is

$$[z^{-2k}]\, R(z)/z^{2h} = \frac{(-c/4)^k}{k!}. \tag{3.2}$$

The factorial growth of the denominator ensures absolute convergence; the alternating sign reflects the negativity of the exponent $-c/(4z^2)$ for $c > 0$. For $c < 0$ the signs become uniform and the R-matrix grows exponentially as $z \to 0$.

4 The c-independent identity $S_3(\mathrm{Vir}_c) = 2$

The cubic shadow coefficient S_3 measures the first higher obstruction beyond the modular characteristic κ.

Theorem 4.1 (c-independence of S_3)**.** *The cubic shadow coefficient of the Virasoro algebra satisfies*

$$S_3(\mathrm{Vir}_c) = 2 \tag{4.1}$$

for all values of the central charge c.

Proof. The cubic shadow coefficient is the BPZ-normalized ratio of the linear T-mode contribution to the constant central-term contribution in the depth-2 collision residue:

$$S_3 \;=\; \frac{(T_{(1)}T)_{\mathrm{lin}}}{(T_{(3)}T)_{\mathrm{const}}}, \qquad \text{read in the Belavin–Polyakov–Zamolodchikov normalization} \tag{4.2}$$

where the three-point function $\langle T(z_1)T(z_2)T(z_3)\rangle$ is normalized against the two-point central term. From the OPE (1.2), the relevant modes are $T_{(3)}T = c/2$ (the central term) and $T_{(1)}T = 2T$ (the linear term, a *field* not a c-number). In the BPZ normalization one replaces the field $2T$ by its expectation on the primary vacuum, which reduces to the scalar $2\kappa(\mathrm{Vir}_c) = c$; the shadow coefficient then reads

$$S_3 \;=\; \frac{2\kappa(\mathrm{Vir}_c)}{\kappa(\mathrm{Vir}_c)} \;=\; 2.$$

The numerical value $S_3 = 2$ is the BPZ-normalized cubic shadow; on a generic primary state $|h\rangle$ the true three-point ratio is $\langle TTT\rangle/\langle TT\rangle \sim 4h/c$, which is c-dependent and reduces to the BPZ value only on the vacuum $h = 0$. The statement $S_3(\mathrm{Vir}_c) = 2$ *for all* c is therefore a theorem about the

BPZ-normalized shadow, not a statement about matrix elements on arbitrary primary states. This normalization is the correct one for the cubic shadow invariant (Definition **??**), since the shadow tower is defined on the vacuum module by construction. □

Corollary 4.2 (Class M membership)**.** *The identity $S_3(\mathrm{Vir}_c) = 2 \neq 0$ certifies that the Virasoro algebra belongs to shadow class M (infinite shadow depth) at every central charge. The critical discriminant $\Delta_{\mathrm{Vir}} = 8\kappa S_4 = 40/(5c+22) \neq 0$ for generic c, and the single-line dichotomy forces $r_{\max} = \infty$.*

5 Non-formality: the class M signature at genus 0

The closed form $R(z) = z^{2h}\exp(-c/(4z^2))$ encodes the genus-0 non-formality of the Virasoro algebra. The first correction beyond the leading power is

$$R_2 = -\frac{c}{4}, \tag{5.1}$$

which is the coefficient from which the infinite primitive-cumulant series $\Delta_{\mathrm{Vir}}(t) = t^3 + 2t^5 + \cdots$ of the Virasoro shadow obstruction tower descends.

For a chirally Koszul algebra, the A_∞ operations m_n on the bar cohomology $H^*(\bar{B}^{\mathrm{ch}}(\mathcal{A}))$ encode the non-formality: $m_n = 0$ for $n \geq 3$ is equivalent to formality (A_∞ quasi-isomorphism to the strict associative algebra). The non-vanishing of R_2 is the genus-0 obstruction to formality.

Proposition 5.1 (Non-formality witness)**.** *Let $\mathcal{A}$ be a chirally Koszul algebra with spectral R-matrix $R(z) = z^{2h}(1+R_2z^{-2}+R_4z^{-4}+\cdots)$ on a primary line. If $R_2 \neq 0$, then $\mathcal{A}$ is not A_∞-formal on that primary line: the transferred A_∞ operation $m_3 \neq 0$.*

Proof. The coefficient R_2 is determined by the modular characteristic: on a primary line, $R_2 = -\kappa/2$, which equals $-c/4$ for Virasoro (where $\kappa = c/2$). The shadow-formality identification at degree 3 gives $S_3 \neq 0 \Leftrightarrow m_3 \neq 0$. Since $R_2 \neq 0 \Leftrightarrow S_3 \neq 0$, the non-vanishing of R_2 witnesses the failure of A_∞-formality. □

For the Virasoro algebra, $R_2 = -c/4 \neq 0$ for all $c \neq 0$: the Virasoro is non-formal at every nonzero central charge. The non-formality is encoded in the single number $c/4$ and propagates through the exponential to produce infinitely many nonvanishing corrections.

6 Comparison with Heisenberg and affine Kac–Moody

The spectral R-matrix provides a sharp invariant distinguishing the three main families.

Example 6.1 (Heisenberg: class G)**.** The Heisenberg algebra $\mathcal{H}_k$ has OPE $J(z)J(w) \sim k/(z-w)^2$. The collision residue is $r(z) = k/z$, a simple pole. On the primary sector, $R(z) = z^k$: the R-matrix is a monomial. There is no exponential correction; $R_2 = 0$ and $S_3 = 0$. The tower terminates at degree 2: class G.

Example 6.2 (Affine Kac–Moody: class L)**.** For $\widehat{\mathfrak{g}}_k$ with Casimir at level k, the collision residue is $r(z) = k\,\Omega_{\mathrm{tr}}/z$ in the trace convention and $r(z) = \Omega_{\mathrm{KZ}}/((k+h^\vee)z)$ in the Kazhdan–Lusztig convention, with $\Omega_{\mathrm{KZ}} = 2h^\vee\,\Omega_{\mathrm{tr}}$; the two conventions differ by a level-dependent rescaling of the Casimir, not by an identity in k. In the trace convention, $r\,|_{k=0} = 0$, while in the KZ convention $r\,|_{k=0} = \Omega_{\mathrm{KZ}}/(h^\vee z)$ remains finite and nonzero (the Lie bracket persists at abelian level). The R-matrix, in the KZ convention used throughout this table for uniformity across families, is $R(z) = z^{\Omega/(k+h^\vee)}$, a monomial in z with the Casimir eigenvalue as exponent. The cubic shadow $S_3 \neq 0$ (from the structure constants $f^{ab}{}_c$), so the tower extends to degree 3; the quartic and higher terms vanish by independent sum factorization: class L.

Example 6.3 (Virasoro: class M)**.** The R-matrix $R(z) = z^{2h}\exp(-c/(4z^2))$ has infinitely many nonvanishing coefficients. The essential singularity at $z = 0$ is the datum that no finite truncation of the shadow obstruction tower captures the full R-matrix: every degree contributes, and the tower does not terminate.

The comparison is summarized in the following table:

Family	$R(z)$	S_3	$r_{\max}$	**Class**
Heisenberg $\mathcal{H}_k$	z^k	0	2	G
Affine $\widehat{\mathfrak{g}}_k$	$z^{\Omega/(k+h^\vee)}$	$\neq 0$	3	L
Virasoro Vir_c	$z^{2h}\exp(-c/(4z^2))$	2	∞	M

The progression from monomial (class G) through monomial with nontrivial exponent (class L) to essential singularity (class M) mirrors the increasing complexity of the OPE: quadratic, Lie-type, and non-quadratic respectively.

7 HZ-1 audit: level prefix, $k = 0$ check, convention bridge

The Virasoro collision residue

$$r^{\mathrm{Vir}}(z) \;=\; \frac{c/2}{z^3} \;+\; \frac{2T}{z}$$

carries the central charge c as explicit level prefix. The HZ-1 audit is: substitute $c = 0$. Both terms vanish: the $c/2$ central term and the T insertion (with $T \equiv 0$ on the $c = 0$ triviality). The $c = 0$ check passes: $r^{\mathrm{Vir}}(z)|_{c=0} = 0$, as required for any trace-form r-matrix.

The *cubic + simple* pole profile is the correct one (AP19, AP21): $d\log$-absorption sends the fourth-order OPE pole $c/2 \cdot (z-w)^{-4}$ to the third-order r-matrix pole $(c/2)/z^3$, and the second-order OPE pole $2T \cdot (z-w)^{-2}$ to the simple r-matrix pole $2T/z$. The simple OPE pole $\partial T/(z-w)$ is absorbed entirely and does not contribute. The common mistakes $r^{\mathrm{Vir}}(z) = (c/2)/z^4$ (AP19, fourth-order, un-absorbed) and $r^{\mathrm{Vir}}(z) = (c/2)/z^2$ (AP27, double pole, over-absorbed) are both excluded by this audit.

Remark 7.1 (Trace-form vs λ-bracket convention bridge). Two conventions for the Virasoro r-matrix coexist in the programme:

(i) Trace-form (Vol I): $r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$.

(ii) λ-bracket (Vol II): $\{T_\lambda T\} = (\lambda^3/6) \cdot c + 2\lambda T + T'$, equivalent to writing the cubic coefficient as $c/12$ in divided-power convention.

The bridge is $S_2^{\mathrm{Vol\,I}} = \kappa^{\mathrm{Vir}} = c/2 = 6S_2^{\lambda\text{-bracket}}$ (AP180): the λ-bracket $c/12$ coefficient is related to the Vol I $c/2$ shadow invariant by the divided-power factor $3! = 6$. Bare $S_2 = c/12$ is the λ-bracket coefficient; the shadow invariant in Vol I notation is $S_2 = c/2$ (AP177).

Remark 7.2 (Connection to κ and the shadow tower). The simple-pole coefficient $2T$ of $r^{\mathrm{Vir}}(z)$ averages (under av) to $2 \cdot \kappa(\mathrm{Vir}_c) = c$ on the primary vacuum (AP177: $\kappa^{\mathrm{Vir}} = c/2$, so $2\kappa = c$). The cubic-pole coefficient $c/2$ is already a scalar: $\mathrm{av}(c/2) = c/2 = \kappa(\mathrm{Vir}_c)$. Both projections recover the modular characteristic, confirming the $E_1 \to$ shadow consistency: the cubic and simple poles project to the *same* κ scalar up to the factor 2 coming from the linear-in-T structure of the simple pole.

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